GroupBuy Dubai: Using Data Analytics to Diagnose and Formulate a Business Strategy

Data Analytics Assignment | Syamantak Dey | SP Jain Global, Dubai

Project repository: https://github.com/babandey648-ux/groupbuy-dashboarb1.git

Draft for review. Verify the results, rewrite the explanations in your own words, complete the reflection in Section 9, and delete this note before submitting.

1. Business strategy and objective

GroupBuy is a proposed Dubai platform that pools demand before asking suppliers to quote. A campaign publishes a product and target quantity, buyers make conditional commitments, suppliers bid against the pooled quantity, and buyers approve the final delivered price before paying. GroupBuy earns an assumed 7% commission on fulfilled orders, with variable cost assumed at 2% of GMV and fixed cost at AED 15,000 a month.

The pitch deck states that the largest early risks are trust and unit economics. The business objective is to decide whom GroupBuy should serve first, what it should sell, how it should make the promised savings credible, and whether the model can pay for itself. No real transactions exist yet, so synthetic data is used to describe and diagnose the whole strategy. The focus is on which data is required, why it is required, how algorithms are applied, how their results are checked, and whether they enable something meaningful. Four questions guide the work:

- Who shows genuine intent, and where do prospects drop out?
- Which categories and features should the first campaigns use?
- Does the promised discount survive delivery costs and meet buyer expectations?
- Can the model reach break-even, and what would stop it?

2. Synthetic data generation

2.1 Why synthetic data, and how it maps to the pipeline

A start-up cannot observe a sales pipeline before it operates. Synthetic data allows the pipeline, its metrics and its decision rules to be built and stress-tested now, so real pilot data can replace it later without redesigning the analysis. The data was designed stage by stage, following the deck's process.

Pipeline stage	Dataset	Key measures	Decision supported
Validate demand	customers.csv (mock survey)	Intent, categories, minimum discount	Is there a reachable audience?
Launch campaign	products.csv, campaigns.csv	Reference price, target quantity, duration	What should be launched?
Aggregate demand	transactions.csv	Committed buyers and value	Was the target met?
Negotiate	suppliers.csv, campaigns.csv	Bids, delivered price, net discount	Is the offer acceptable?
Confirm and pay	transactions.csv	Paid flag, GMV, commission	Did commitment become revenue?
Fulfil and retain	campaigns.csv, transactions.csv	On-time delivery, disputes, repeat buyers	Should we scale?

2.2 Datasets

Table	Rows	Columns	Source and role
Raw mock survey	1,215	63	Respondent profile, attitudes, category and feature choices, intent

Table	Rows	Columns	Source and role
customers	1,200	64	Cleaned survey; one row per buyer
products	48	6	Synthetic: 12 categories, 4 generic items each
suppliers	24	5	Synthetic: fictional names, reliability, price position
campaigns	120	18	Synthetic: Year 1 campaigns (January to December 2027)
transactions	4,449	14	Synthetic: one row per buyer commitment

2.3 How the data was generated logically

The operating tables are linked to the survey through customer IDs, so they are not independent random numbers. The simulation used a fixed random seed, so it can be reproduced exactly.

- Campaign categories were drawn in proportion to survey category interest.
- Only buyers interested in a category could commit, and their quantity came from their stated order value.
- Net discount rises with pooled value and supplier bids, is capped by category, and is reduced by delivery cost of 1.2% to 4% of price.
- Campaigns are cancelled when pledges fall below 70% of target, net discount is under 5%, or paid quantity falls below 40% of target. This applies the deck's Trust Rule.
- A buyer's probability of paying rises when the discount exceeds their stated minimum, and falls when the campaign runs longer than their maximum wait.

Integrity checks confirmed that every customer, campaign, product and supplier key resolves, and that GMV reconciles between campaigns and transactions. Commission, cost and cancellation rules come from the deck; the 120-campaign volume, item prices, discount ceilings, reliability and payment behaviour are my assumptions. The survey file is labelled mock, so neither source is evidence of real demand.

2.4 Which columns are required, and why

The 107 columns are documented individually in Dataset/Data_Dictionary.csv. The main groups are summarised here.

Column group	Why it is required	Used in
Buyer profile (income, area, SME size)	Identifies who buys and where demand can be pooled	EDA, clustering, regression
Current behaviour (spend, orders, channel)	Shows the purchases available to capture	Clustering, classification, regression
Attitudes (price, wait, trust, convenience)	Tests fit with a slow, pooled, trust-based model	Correlation, clustering, classification
Category selections	Shows what can be pooled together	Apriori, campaign simulation
Feature selections	Shows what must be built first	Apriori, classification
Terms (minimum discount, maximum wait)	Sets the offer buyers will accept	Diagnostics, classification
Intent (join, register, contact)	Measures the outcome; never used as a predictor	Classification target, funnel
Expected order value	Sizes campaigns and supplier requests	Regression target
Prices, bids, net discount	Verifies whether savings are real	Diagnostics, unit economics
Paid, GMV, commission, cost	Tests whether commitments create revenue	Sales analysis, break-even
On-time, dispute	Tracks the deck's fulfilment KPIs	Operations review

3. Data preparation

3.1 Cleaning

The raw survey contained deliberate quality problems. Every rule was logged. Implausible values were set to missing instead of being guessed, and later imputation in modelling uses training data only, which prevents leakage.

Issue	Column	Rows	Treatment
Duplicate submissions	Respondent ID	15	Kept first
Text or error spend ("AED 1,310", -1)	Monthly eligible spend	23	Parsed; -1 missing
Age outside 18 to 80	Age	4	Set missing
Household above 10	Household size	6	Set missing
Income outside AED 2,000 to 150,000	Income	13	Set missing
Orders above 40 a month	Online orders	5	Set missing
Discount above 60%	Minimum discount	6	Set missing
Likelihood outside 0 to 10	Join likelihood	4	Set missing
Inconsistent labels	Gender	79	Standardised
Inconsistent labels	Area	42	Standardised
Inconsistent labels	Primary channel	33	Standardised
Inconsistent labels	Pilot registration	82	Standardised

SME employee count was blank for individuals by design, so it was set to 0. Monthly spend still contains 10 values above AED 30,000. None come from SMEs, so they may be entry errors, but they could not be confirmed either way. They were retained and flagged, and their influence is reduced by a log transformation and rank-based correlation.

3.2 Transformation

New variable	Method	Purpose
High_Intent	1 if conversion segment is High	Clear diagnostic outcome
Log_Spend, Log_Income	log(1 + value)	Reduce right skew
Income_Band	Four bands	Compare intent by income
Prior_Group_Buying	Ordinal 0, 1, 2	Use order of experience
WhatsApp_Group, Shared_Contact	Binary 0, 1	Numeric analysis
Category_Count, Feature_Count	Sum of selections	Breadth of interest
Target_Attainment	Pledged ÷ target quantity	Campaign viability
Paid_Share	Paid ÷ committed buyers	Offer conversion
Month	From start date	Time-series aggregation

4. Descriptive analytics: what the data shows

Variable	Count	Mean	Median	Min	Max
Age	1,172	36.7	37	18	60
Monthly income (AED)	1,081	24,208	18,700	5,400	108,700
Monthly eligible spend (AED)	1,197	4,042	2,510	0	110,358
Online orders per month	1,159	6.7	6	0	17
Minimum discount to switch (%)	1,194	16.7	16	2	38
Maximum wait (days)	1,164	5.4	5	0	15

Variable	Count	Mean	Median	Min	Max
Expected order value (AED)	1,200	726	490	50	6,320

Mean spend and order value sit far above their medians, so a few large buyers pull the averages up. Medians are the better guide when planning a typical campaign.

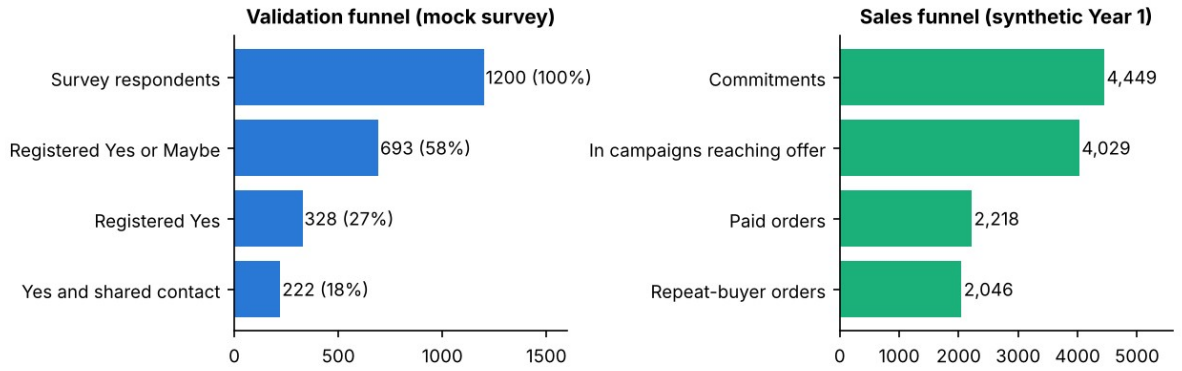


Figure 1. Validation and sales funnels

What it shows. Of 1,200 respondents, 693 (58%) said Yes or Maybe to the pilot, 328 (27%) said Yes, and 222 (19%) also shared contact details. In the simulated sales funnel, 4,449 commitments became 2,218 paid orders, or 50%.

Interpretation. The largest loss occurs between general interest and a firm yes. Sharing contact details is a stronger signal than a stated intention, so 222 is the realistic lead pool. It comfortably exceeds the deck's 30-commitment pilot target. The 50% paid share meets the deck's target only under simulated assumptions.

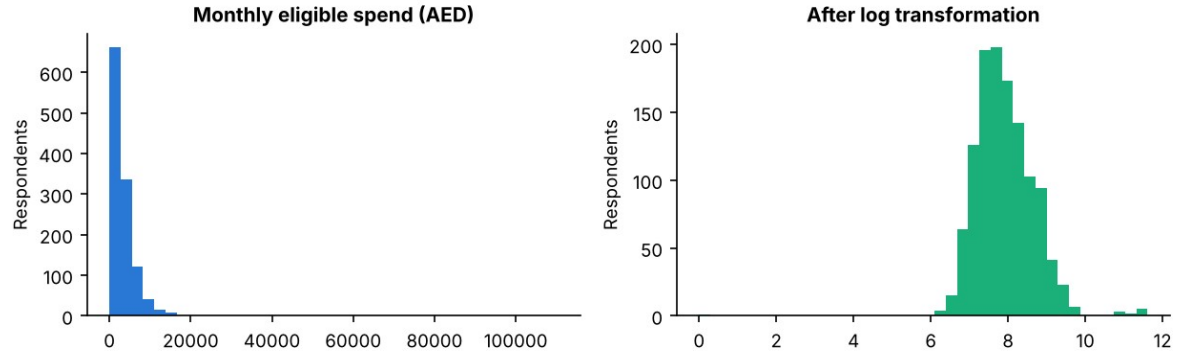


Figure 2. Monthly eligible spend before and after log transformation

What it shows. Raw spend is extremely right-skewed (skewness 10.09), with most buyers compressed at the left. After log transformation, skewness is -0.05 and the distribution is nearly symmetrical.

Interpretation. The transformation justifies the preparation step: without it, a few very large buyers would dominate distance-based methods and linear measures.

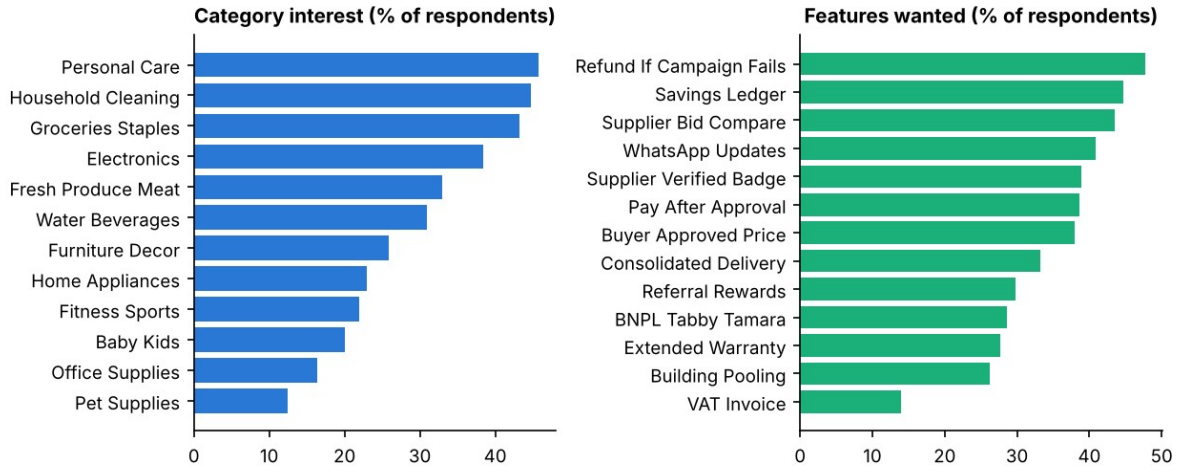


Figure 3. Category interest and desired platform features

What it shows. Personal care (45.8%), household cleaning (44.8%) and groceries (43.2%) lead category interest, whereas office supplies (16.3%) and pet supplies (12.5%) are niche. Refund if a campaign fails (47.7%), savings ledger (44.8%) and supplier bid comparison (43.5%) are the most wanted features.

Interpretation. The first campaigns should use frequently bought household goods. The top features all reduce uncertainty, which supports the deck's view that trust is a central risk.

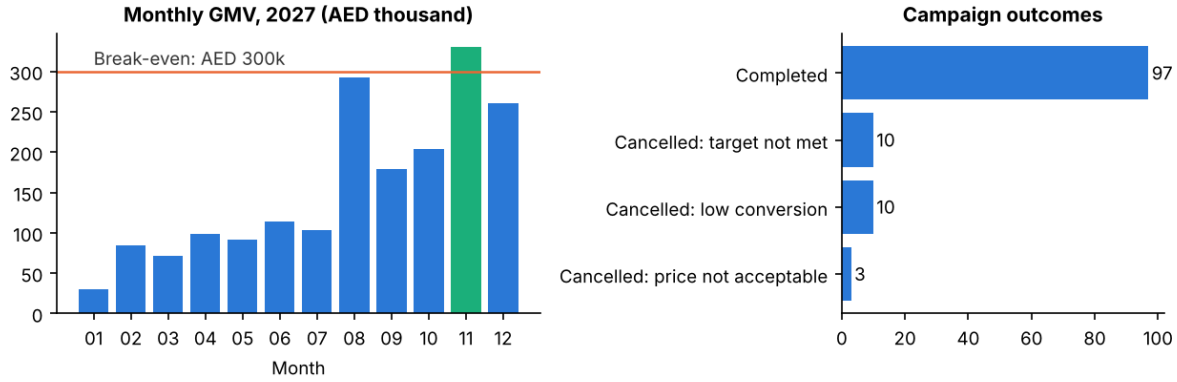


Figure 4. Monthly GMV against break-even, and campaign outcomes

What it shows. Simulated GMV totals AED 1.86M, climbing from AED 30,669 in January to a peak of AED 330,320 in November. Only November clears the AED 300,000 break-even line (AED 15,000 ÷ 5% contribution). Of 120 campaigns, 97 completed, 10 missed target, 10 had low conversion and 3 had an unacceptable price.

Interpretation. Commission of AED 130,331 less variable cost of AED 37,238 leaves AED 93,094, which is AED 86,906 short of annual fixed costs. This reveals a tension in the deck: its break-even implies AED 3.6M annual GMV, but its Year 1 forecast is AED 1.2M.

5. Diagnostic analytics: why it happens

Spearman rank correlation was used because many variables are 1 to 5 ratings and spend is skewed. Correlation shows association only, not cause.

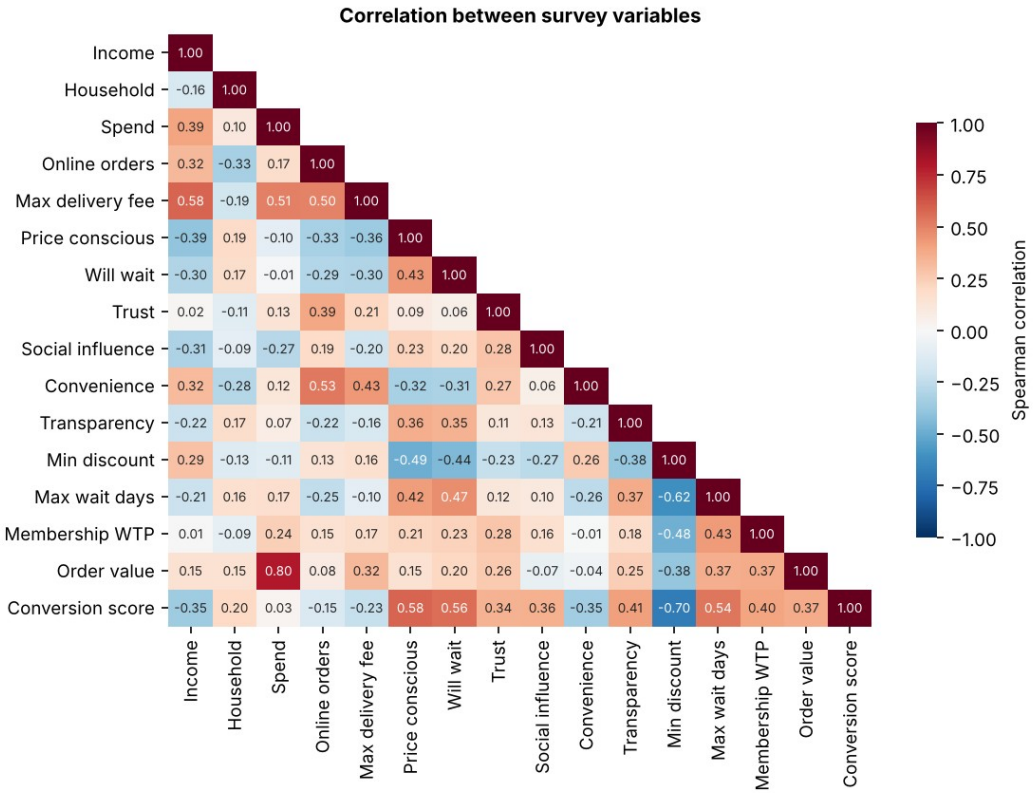


Figure 5. Spearman correlation matrix of survey variables

What it shows. Strongest pairs: spend and order value (0.80), minimum discount and conversion score (-0.70), and minimum discount and maximum wait days (-0.62). Income relates positively to the delivery fee buyers accept (0.58). Online orders relate to convenience priority (0.53).

Interpretation. Two groups emerge. Bargain-oriented buyers accept smaller discounts and longer waits. Affluent, convenience-oriented buyers order often, tolerate delivery fees and want a larger incentive to change. GroupBuy's slow, pooled model naturally suits the first group.

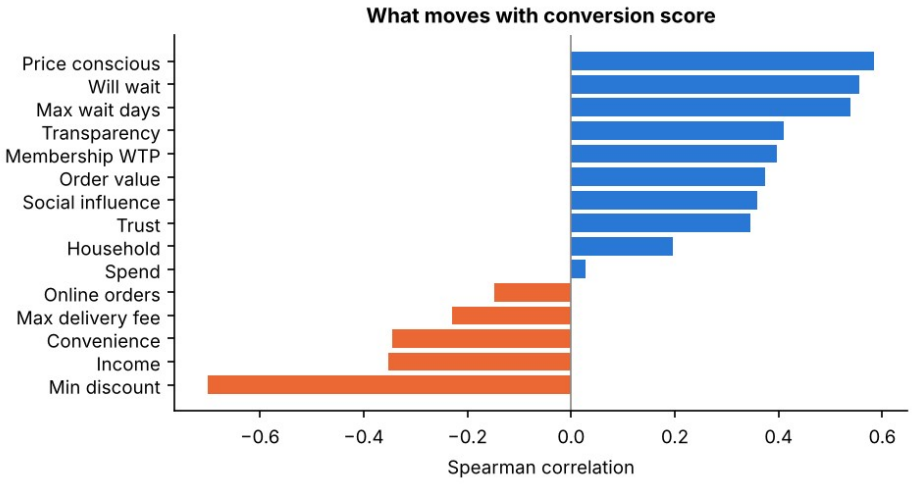


Figure 6. Variables most associated with conversion score

What it shows. Conversion score falls as the required minimum discount (-0.70), income (-0.35) and convenience priority (-0.35) rise. It rises with price consciousness (0.58), willingness to wait (0.56), maximum wait days (0.54) and transparency (0.41). Spend has almost no relationship (0.03).

Interpretation. Attitude predicts conversion better than purchasing power. Targeting high spenders alone would be a mistake. Marketing should screen for price sensitivity and patience.

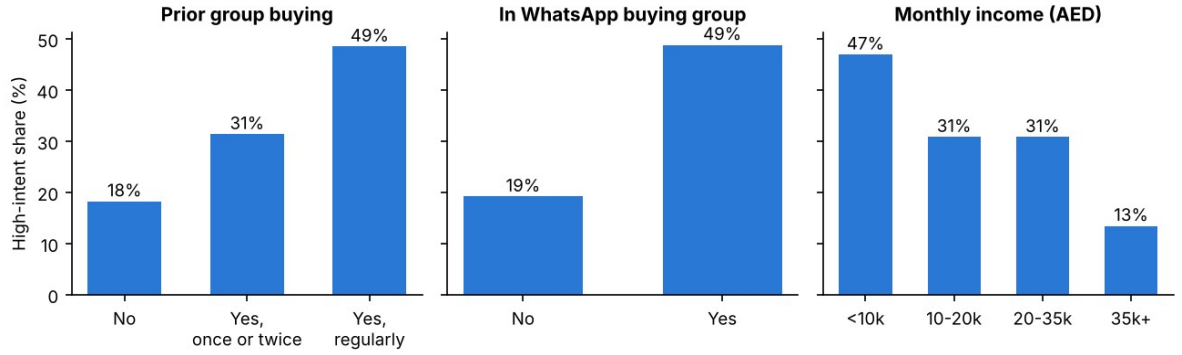


Figure 7. High-intent share by experience, community and income

What it shows. High-intent share climbs from 18% for those with no group-buying experience to 49% for regular users. WhatsApp buying group members reach 49%, against 19% for non-members. Income shows a reverse pattern, from 47% below AED 10,000 to 13% above AED 35,000.

Interpretation. Existing WhatsApp groups are an efficient starting channel, consistent with the deck's manual WhatsApp pilot. The lowest income band has only 98 respondents, so its figure is less stable.



Figure 8. Discount buyers require compared with net discount offered

What it shows. The median buyer needs 16%, but the median simulated net discount is 13.0%. Only 30.5% of respondents would accept the median offer on price alone.

Interpretation. This diagnoses the deck's "savings disappear in delivery" risk. Campaigns need larger pooled volume, better supplier terms or lower delivery cost before the headline offer is competitive.

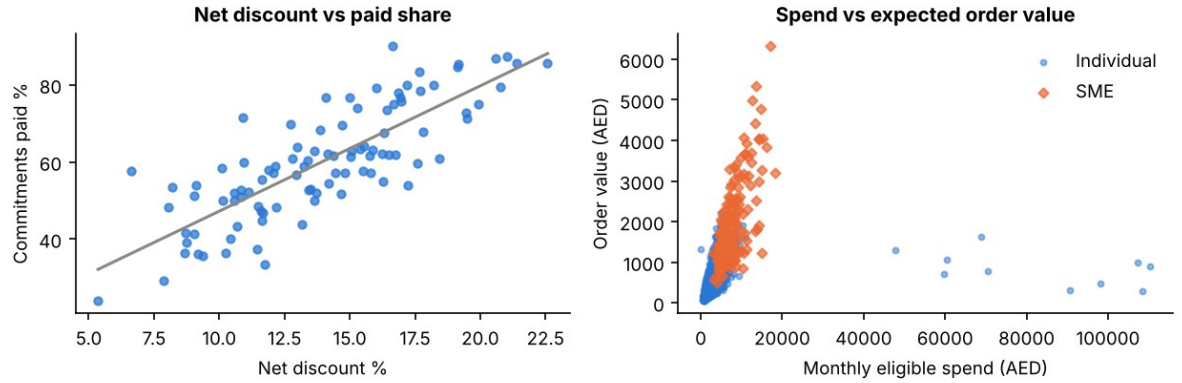


Figure 9. Net discount against paid share, and spend against order value

What it shows. Across 97 completed campaigns, net discount and paid share correlate at 0.81. Paid share rises from 42% below a 10% discount to 75% at 16% or more, about 3.25 percentage points for each discount point. In the right panel, order value rises with spend for individuals (0.73) and SMEs (0.60). Median SME order value is AED 1,750, against AED 430 for individuals.

Interpretation. The left panel partly reflects a rule built into the simulation, so it confirms the pipeline logic instead of proving real price response. A real pilot should measure this slope. The right panel supports separate campaign tracks and forecasts for SMEs.

Why campaigns failed

Target-not-met campaigns averaged 61% of target quantity and a 10.7% net discount. Low-conversion campaigns reached 88% of target but offered 9.5%. Completed campaigns averaged 161% of target and 13.9%. Failure therefore has two separate causes: too little demand, or enough commitments but an offer too weak to convert them.

6. Analytics techniques: rationale and results

Each technique was chosen for a specific business decision. All predictive models exclude the intent answers and conversion score, because the score is almost exactly a weighted sum of those answers (R^2 about 0.99). Using them would give impressive but meaningless accuracy.

Technique	Why this technique	Input	Business output
K-Means clustering	No segment labels exist; groups similar buyers	13 scaled behaviours	Target segments
Classification	The outcome is yes or no: high intent or not	52 non-intent fields	Lead priority
Apriori	Finds items that occur together in baskets	12 categories, 13 features	Bundles and offers
Regression	The outcome is a number in AED	17 non-intent fields	Campaign size

6.1 Clustering

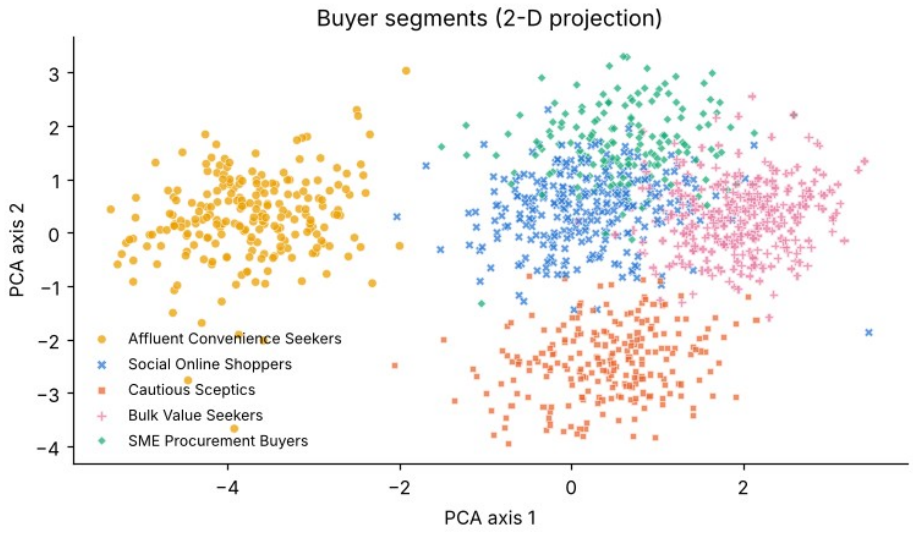


Figure 10. K-Means buyer segments, shown in two dimensions

Five clusters were selected because k=5 is the elbow and a local silhouette peak. Two clusters scored higher (0.296 against 0.237) but only separated affluent buyers from everyone else. The five segments differ sharply in intent, although intent was not used to create them: Bulk Value Seekers 59%, SME Procurement Buyers 45%, Social Online Shoppers 22%, Cautious Sceptics 4% and Affluent Convenience Seekers 4%. SME buyers have a median expected order of AED 1,720.

6.2 Classification

Test-set model	Accuracy	Precision	Recall	F1
Always majority (baseline)	72.3%	0%	0%	0.000

Test-set model	Accuracy	Precision	Recall	F1
Logistic Regression	81.3%	63.1%	78.3%	0.699
Decision Tree	75.7%	54.3%	75.9%	0.633
Random Forest (selected)	77.7%	56.6%	83.1%	0.673

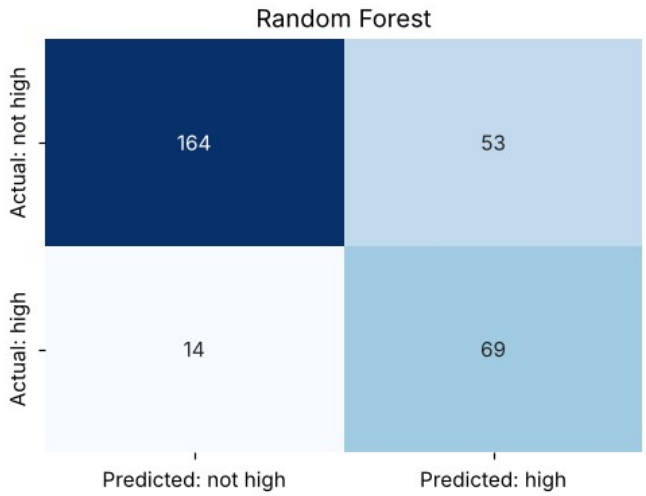


Figure 11. Confusion matrix for the selected Random Forest

On 300 test leads, 69 true positives are high-intent leads correctly called. The 164 true negatives are correctly left in low-cost nurture. The 53 false positives are wasted calls, and the 14 false negatives are missed buyers. Accuracy alone would mislead, because a model that predicts nobody reaches 72.3% and finds no customers. In cold start a missed buyer costs more than a wasted message, so recall is the priority.

6.3 Association rule mining

If a buyer wants	They also want	Support	Confidence	Lift
Baby + Groceries	Fresh produce and meat	11.2%	87%	2.64
Baby	Fresh produce + Groceries	11.2%	56%	2.72
Building pooling + WhatsApp updates	Consolidated delivery	11.4%	86%	2.59
Buyer-approved price + Pay after approval	Refund if campaign fails	14.2%	89%	1.87

Support is how common the combination is, confidence is how reliable the rule is, and lift shows how much stronger it is than chance. These rules support family bundles on one delivery run, a packaged building campaign, and a joint trust guarantee.

6.4 Regression

Test-set model	MAE (AED)	MSE	RMSE (AED)	R^2
Mean baseline	452	405,529	637	-0.00
Multiple Linear Regression	322	214,297	463	0.47
Linear Regression (log target)	298	215,942	465	0.47
Random Forest Regressor (selected)	225	117,008	342	0.71

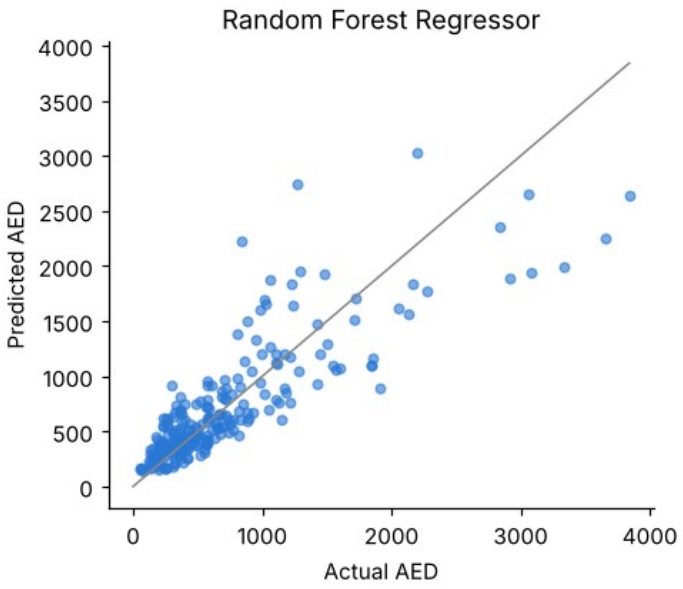


Figure 12. Actual and predicted expected order value

The closer points are to the diagonal, the better the prediction. Eligible spend is the main driver. Mean error is AED 164 for individuals but AED 609 for SMEs, so SME demand needs separate planning.

7. Are the results right, and are they meaningful?

A good score on one split is not proof. Each technique was therefore tested again across repeated samples.

Technique	Test	Result
Classification	5-fold cross-validation, 3 repeats	Random Forest F1 0.673 (SD 0.043), ROC-AUC 0.867; Logistic Regression F1 0.667
Classification	Labels shuffled to detect leakage	ROC-AUC 0.49, about the same as guessing
Regression	5-fold cross-validation, 3 repeats	Random Forest R^2 0.738 (SD 0.030), MAE AED 224; Linear R^2 0.529
Clustering	Refit on 30 random 80% samples	Adjusted Rand Index 0.990: segments are stable
Apriori	Mined on two separate halves	62 of 65 half-A rules also appear in half B; lift correlation 0.95

Random Forest and Logistic Regression are effectively tied on F1, given the variation between folds. A founder might reasonably prefer Logistic Regression because it is simpler to explain. The shuffled-label test falling to chance indicates that the pipeline is not leaking the answer. These tests show that the methods are internally sound; they cannot show that mock answers match real buyers.

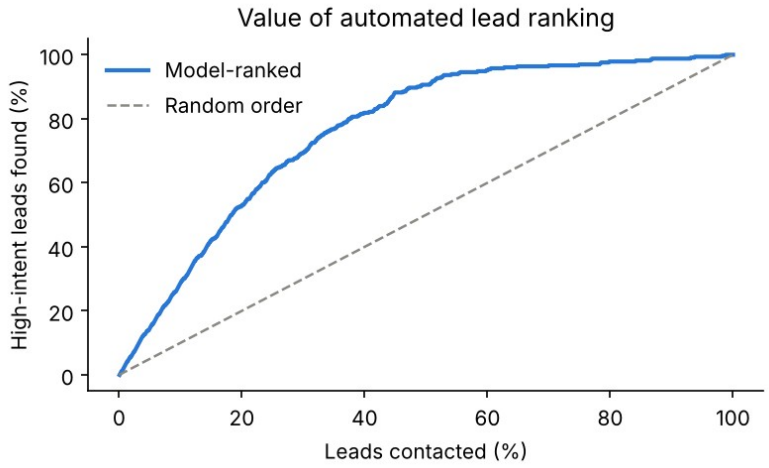


Figure 13. Share of high-intent leads found by model-ranked and random contact order

Is the automation useful? Yes, within this dataset. If GroupBuy can contact only 20% of its 1,200 leads, ranking by model score finds 175 high-intent leads, compared with about 66 in random order, or 2.6 times as many. At 50% it finds 91% of them.

Campaign planning. For 1,000 simulated 40-buyer campaigns, the regression forecast has a median error of 5.3% of campaign value, about half the 10.6% error from using the overall average. That makes target quantities sent to suppliers more dependable.

Technique	Decision	Automation
Clustering	Which segment gets which offer	Assign each sign-up to a segment message flow
Classification	Who the founders call first	Route score to call list or WhatsApp nurture
Apriori	What to bundle	Suggest the highest-lift open campaign at commitment
Regression	How much demand to promise	Calculate suggested target quantity

8. Streamlit dashboard

The same data and model outputs feed an 11-section Streamlit dashboard covering performance, customers, correlation, all four techniques, validation and recommendations. The screenshots below are actual captures of the running application.

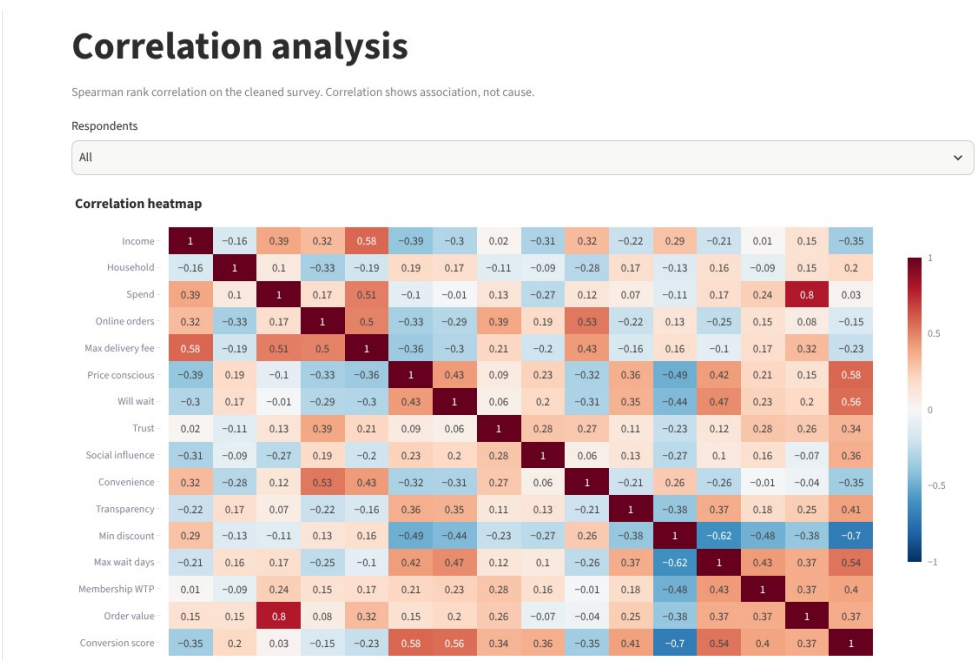


Figure 14. Streamlit dashboard: correlation analysis

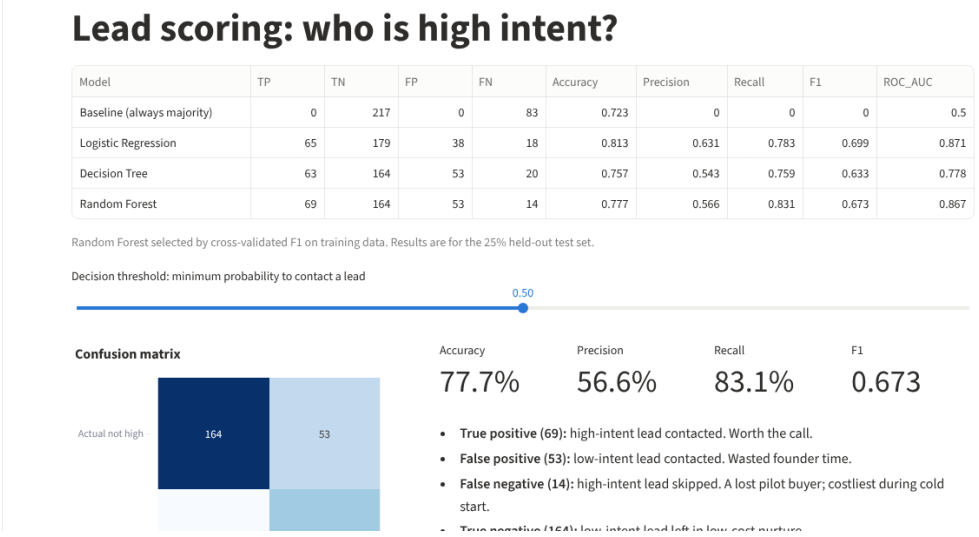


Figure 15. Streamlit dashboard: classification with adjustable threshold

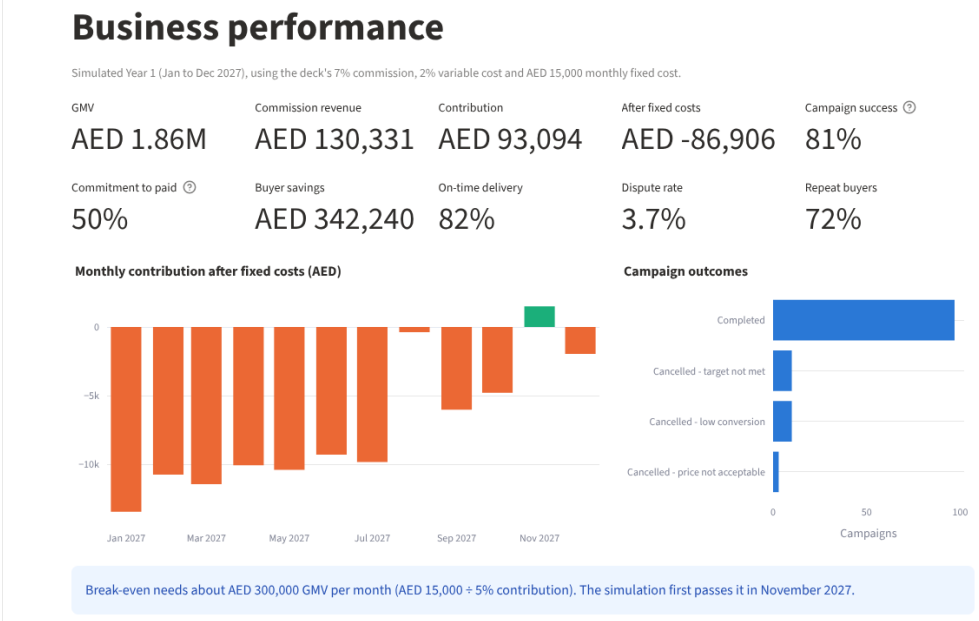


Figure 16. Streamlit dashboard: simulated business performance

9. Recommended strategy and conclusion

Strategy element	Evidence	Formulated decision
Target market	Bulk Value Seekers 59% high intent; WhatsApp members 49% vs 19%	Launch with price-sensitive buyers through existing WhatsApp groups
Product	Household, personal care and groceries lead interest	Start with staples; add family bundles
SME track	Median order AED 1,720, largest simulated GMV segment	Run a separate procurement track
Pricing	Median need 16%; median simulated offer 13.0%	Do not launch offers that fall well short; pool more or cut delivery cost
Trust	Refund guarantee is the top feature at 47.7%	Lead with refund, approval and savings ledger
Economics	Break-even AED 300,000 GMV monthly; simulated loss AED 86,906	Revisit Year 1 plan, volume or fixed costs

The analysis supports a pipeline with explicit gates. Thresholds come from the deck, the survey or the simulation, and should be confirmed in the pilot.

Gate	Proceed when	If not
Lead qualification	Pilot Yes and contact shared	Move to low-cost WhatsApp nurture
Demand	Pledges at least 70% of target	Revise or cancel, with terms disclosed
Offer	Net discount at least 5%; aim for 16%	Seek more bids or consolidate delivery
Conversion	At least 50% of eligible commitments paid	Review discount and campaign duration
Fulfilment	On-time delivery improves on the simulated 82%; disputes stay low	Replace or re-verify the supplier
Scale	Monthly GMV near AED 300,000, with repeat purchases	Modify or stop, as the deck's Day 90 review states

The analysis suggests GroupBuy should launch with price-sensitive, patient buyers already in WhatsApp groups, focusing on household staples with a visible refund guarantee. It should treat SMEs as a separate high-value track, and close the gap between its typical offer and the 16% discount buyers want. Its Year 1 financial plan needs revisiting.

The main limitation is the data. Survey answers are mock and operating records are synthetic, so patterns partly reflect assumptions. The results finalize how to validate the business; they do not prove that the business is valid. The next step is to replace the simulated tables with real data from the 90-day pilot and rerun the same analysis.

[Your reflection] In your own words: which graph changed your view of GroupBuy most, and what did you learn about cleaning data or interpreting correlation?